



Chihyeon Yun

AI Researcher & Engineer

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- ◆ **Multimodal Auto-Editing (M.S. & First Author):** Designed visual speaking-state (Swin Transformer) and audio diarization (WhisperX / Pyannote) late-fusion frameworks, utilizing Beam Search decoding to estimate optimal camera cuts (achieved 90.10% agreement with experts).
- ◆ **Spatial AI & Metaverse (ETRI R&D):** Engineered high-accuracy indoor Visual Positioning Systems (VPS) for cyber-physical space registration, and designed Visual-Language Model (VLM) zero-shot anomaly detection pipelines for real-time video semantics.
- ◆ **Acoustic Surveillance (Scientific Data, Nature SCIE):** Built the first open Military Audio Dataset (MAD, 8,075 samples) for situational awareness, designed log-Mel filterbank feature extractions, and benchmarked Audio Spectrogram Transformers (AST, 91.07% accuracy).
- ◆ **Medical AI & Audio Diffusion (NeurIPS Workshop):** Addressed diagnostic data scarcity via conditional audio diffusion (DiffWave) synthesis and designed Adversarial Fine-Tuning (AFT) models to align domain features (improving minority class classification by up to 26.58%).
- ◆ **3D Reconstruction & VLA Robotics (PoC):** Coupled 2D MT-DUNet (DCNv2) and Beer-Lambert radiative 3DGS for coronary artery 3D reconstruction under extreme memory optimization, and integrated SmolVLA control in Unity ROS simulation environments.
- ◆ **Multimodal Emotion Recognition (KCC & ETRI Award):** Integrated text (KLUE-RoBERTa) and speech (wav2vec 2.0) encoders using Latent CutMix augmentation and Supervised Contrastive Learning (SupCon) for cosine similarity alignment (F1-score 71.17%).

📁 Research & Career Timeline

2025.02 - 2025.06

Electronics and Telecommunications Research Institute (ETRI)

Post-Master's Research Fellow at Hyper-Realistic Metaverse Lab. Worked on indoor visual positioning (VPS) coordinates mapping and VLM semantic anomaly detection. Managed joint government R&D milestones.

2023.03 - 2025.02

Kyungpook National University

Master of Science in Artificial Intelligence (GPA: 4.11/4.3). Conducted research in Multimodal Language & Cognition Lab on audio-visual late-fusion, speech diarization, and generative audio diffusion.

🔧 Core Tech Stack & Skills

Deep Learning & Audio

PyTorch Transformers DiffWave Ko-wav2vec 2.0 KLUE-RoBERTa
SupCon WhisperX Pyannote.audio Librosa

Vision & Spatial AI

3DGS Visual SLAM/VPS MT-DUNet (DCNv2) Swin Transformer OpenCV
MediaPipe

VLMs, Robotics & DevOps

VLA / SmolVLA VLM Zero-shot ROS/ROS2 Unity Engine Docker
Pinecone Vercel Git

Medical Imaging

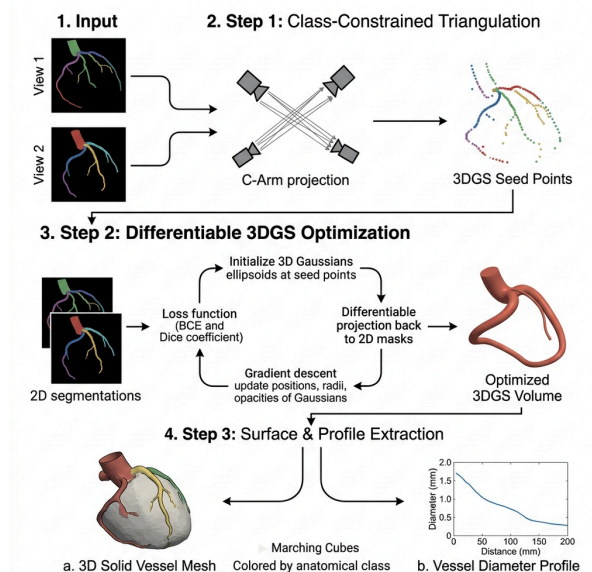
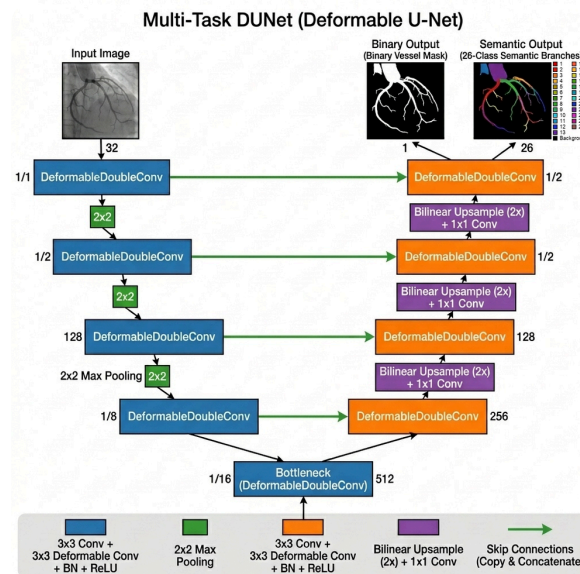
3D Reconstruction

LCA CAG 3DGS Reconstruction & Dashboard

A 3D volumetric reconstruction PoC that combines a 2D Multi-Task Deformable UNet (MT-DUNet) and Beer-Lambert radiative 3D Gaussian Splatting (3DGS) to recover 3D coronary artery structure, radius profile, and volume from sparse 2D C-arm X-ray angiograms.

- ✦ **MT-DUNet (DCNv2):** Integrates deformable convolutions to adapt to complex vessel tortuosity, simultaneously predicting binary vessel masks and 25 anatomical branch labels.
- ✦ **Class-Constrained Triangulation:** Prevents ghost-point artifacts in epipolar matching by pairing identical anatomical branch labels across views, yielding a 4.5x denser centerline seed.
- ✦ **Beer-Lambert Radiative 3DGS:** Solves the transparency occlusion problem of X-ray projection by analytically integrating density along rays, optimizing vessel radius (r) and opacity (α).
- ✦ **High Efficiency:** Achieved 8x GPU memory saving (2.5 GiB execution) on an L4 GPU via downsampling and multi-resolution rendering.

Tech Stack: PyTorch, 3DGS, Deformable Conv (DCNv2), Python, HTML/JS



AI Agent

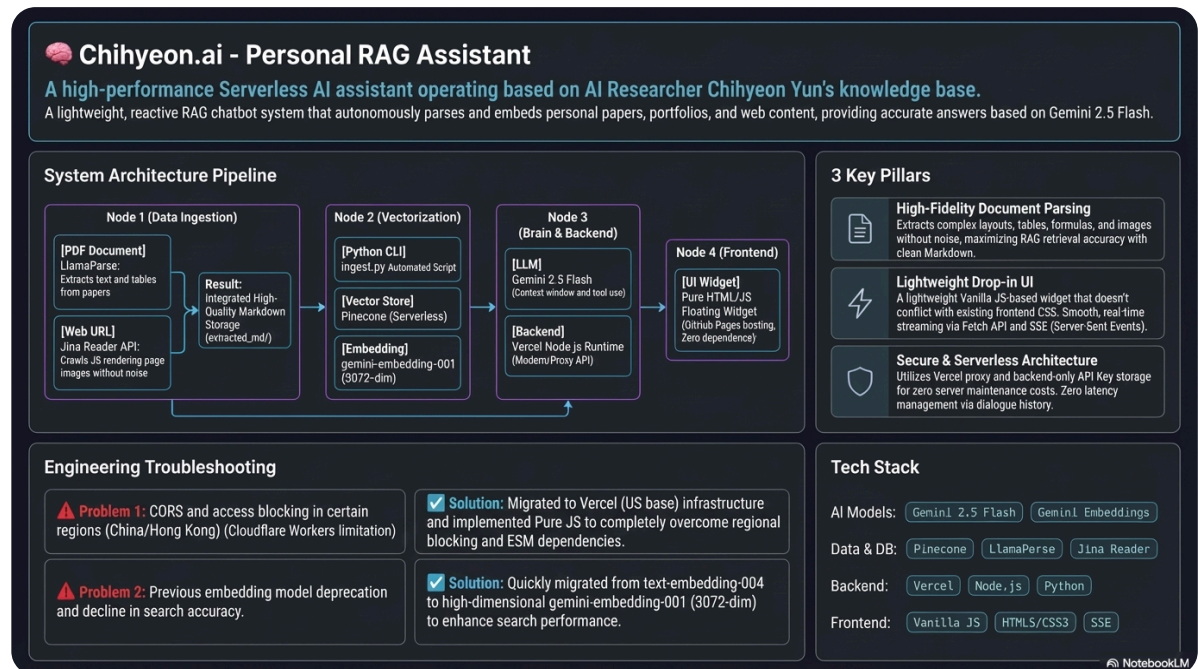
RAG Systems

Chihyeon.ai Personal RAG Assistant

A sophisticated personal RAG system that serves as an interactive AI secretary, providing high-fidelity, grounded answers based on academic research papers, CV, and project documentation.

- ◆ **High-Fidelity Parsing:** Leverages LlamaParse & Jina Reader API to flawlessly extract complex academic layouts, mathematical formulas, and JS-rendered web content.
- ◆ **Secure Serverless Proxy:** Uses Vercel Functions to encrypt API credentials and bypass region locks, ensuring safety and compliance.
- ◆ **Vector Storage:** Fast semantic matching over high-dim Pinecone space powered by Gemini 2.5 Flash for near-instant, streaming answers.

Tech Stack: Gemini 2.5 Flash, Pinecone (Serverless), Vercel Functions, LlamaParse, Jina Reader, Python



Robotics

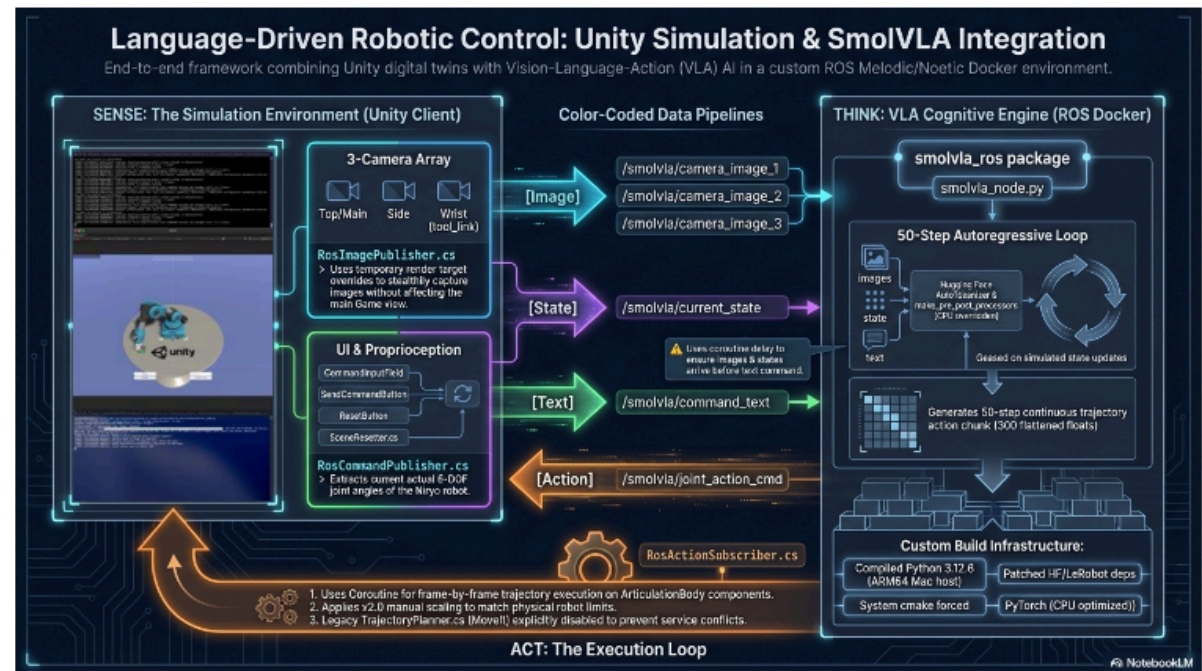
Simulation

Unity Pick & Place SmolVLA Integration

An end-to-end integration of the SmolVLA (Vision-Language-Action) model within a virtual physics environment to perform robotic 'Pick and Place' tasks.

- ◆ **Vibe Coding Paradigm:** Worked with LLMs to prototype network interfacing, setup boilerplate code, allowing focus on core logic.
- ◆ **Communication Pipeline:** Built ROS-Unity bridges linking virtual cameras and joint torques for real-time simulation feedback.
- ◆ **Visual Control Loop:** Feeds camera images directly to SmolVLA agents for high-level spatial coordination and torque control verification.

Tech Stack: SmolVLA, Multimodal LLM, Unity Engine, ROS, Docker, PyTorch



Seamless Multi-Hypothesis Video Editing Using Deep Multimodal Analysis

Fuses active speaker detection (Swin Transformer) and diarization (WhisperX/Pyannote) visual-audio streams to align channels, decoding edits via a customized Beam Search logic matching 90.10% of expert visual pacing.

- Proposed a late-fusion multi-camera synchronization and editing framework, integrating speech state classification and visual cue matching into a Beam Search sequence decoder.

Tech Stack: Speaker State Estimation, Swin Transformer, WhisperX, Pyannote.audio, Beam Search, Python

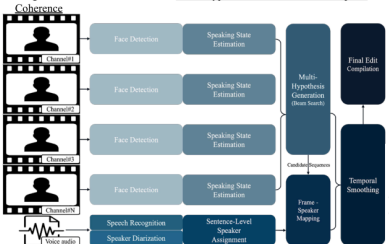
Seamless Multi-Hypothesis Video Editing Using Deep Multimodal Analysis Over Multi-Channel Sources

Introduction

- An automated system for editing multi-channel video by synchronizing raw inputs through separate visual and auditory pipelines.
- Deep late-fusion of multimodal information for face recognition, speaking-state estimation, speech recognition and speaker identification.
- A multi-hypothesis editing approach that provides candidate versions based on probability distribution over time for extracting frames from each channel stream.
- Frame-level match accuracies of 84.99% and 90.10% for the 3-channel and 5 channel videos, highlighting the system's robust performance across varying speaker complexities.

Method

- For multi-speaker, multi-channel content, we adopted an **intuitive approach** to make automated editing feasible. This is primarily because **datasets containing multi-speaker, multi-channel videos paired with their fully edited versions are extremely rare**.
- The lack of such datasets makes it challenging to apply standard supervised learning techniques effectively for automated video editing.
- For the audio modality, our goal is to accurately detect active speech segments and identify the corresponding speakers from diverse and often noisy multi-channel sources. - **Speech Recognition, Speaker Diarization**
- The visual approach involves estimating the speaking state based on the facial region of the speaker. - **Face Detection, Speaking State Estimation**
- Integration and Channel Selection - **Multi-Hypothesis Generation for Temporal Coherence**



Experiment & Result

- Dataset
 - Studio recordings provided by a Korean public broadcasting company in South Korea. {3-channel video dataset, 5-channel video dataset}
- Evaluation
 - Compare the ground-truth labeled data with output generated by our system at the frame level.
 - For the Quantitative Analysis, Analyze the frame-level alignment accuracy under various smoothing and temporal conditions.

	15	30	45
Best Path (self-transition bias 0.7)			
3-channel video	0.8499	0.8495	0.8495
5-channel video	0.8995	0.8999	0.8854
Best Path (self-transition bias 0.9)			
3-channel video	0.8499	0.8497	0.8481
5-channel video	0.8997	0.8910	0.8665

TABLE I: Match rates (Top Beam Paths applying self-transition bias 0.7/0.9) varying the smooth threshold.

Length (s)	3-channel video		5-channel video	
	Match	Rate	Match	Rate
0-10	4,649	4.649	1,779	2.961
10-20	4,482	4.482	1,845	3.487
20-30	5,121	5.121	1,855	3.571
30-40	2,971	3.214	1,855	3.571
40-50	1,486	1.148	1,855	3.571
50-60	3,104	3.315	1,855	3.571
60-70	2,128	2.128	1,855	3.571
70-80	2,128	2.128	1,855	3.571
80-90	2,128	2.128	1,855	3.571
90-100	2,128	2.128	1,855	3.571
Total	22,871	26.406	16,330	18.100

TABLE II: Match rates across different segment lengths.

Criteria	Obs	Exp
1. Speaker Switching Smoothness & Synchronization	4.261101	4.516176
2. Is the system transition smooth when the speaker changes in the video?	4.261101	4.516176
3. Are the cut transitions in the video smooth?	4.261101	4.516176
4. Noise Consistency	4.716108	4.516176
5. Is the noise in the video consistent? Based on the speaker	4.716108	4.516176
6. All Editing Potential	4.716108	4.516176
7. Is the system able to edit professional editing video and generate a high-quality video?	4.716108	4.516176
8. Is the AI system able to edit professional editing video and generate a high-quality video?	4.716108	4.516176

TABLE III: Subjective Video Quality Evaluation Results.

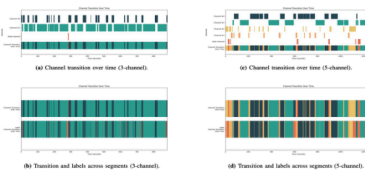
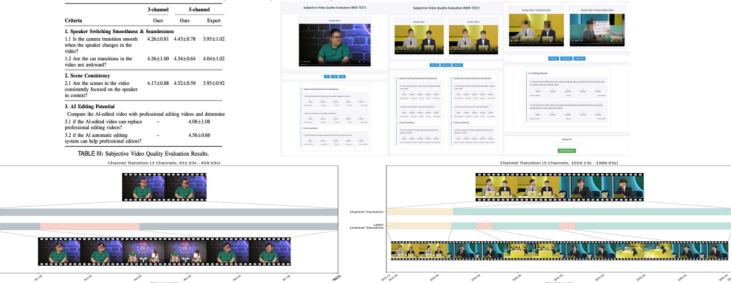


Fig. 2: Visualization of channel selection for the 3-channel (left column) and 5-channel (right column) videos. Each sub-figure shows the overall transitions and a detailed view with ground-truth labels.



02

Scientific Data (Nature Portfolio, SCIE, 2024 / Co-Author)

A Military Audio Dataset for Situational Awareness and Surveillance

Curated and open-sourced 8,075 military audio clips (MAD) representing 7 hazard classes for security surveillance, designing raw audio preprocessing filters and evaluating AST models achieving 91.07% classification accuracy.

- ✦ Built audio feature extraction pipelines (Librosa log-Mel filterbanks) and structured AST-Patch-Mix classifier benchmarks for real-time audio defense surveillance.

Tech Stack: Audio Spectrogram Transformer (AST), Mel Filterbank, Python, Librosa

A Military Audio Dataset for Situational Awareness and Surveillance

Introduction

- Audio classification related to military activities is a challenging task due to the high levels of background noise and the lack of suitable and publicly available datasets.
- To bridge this gap, this paper constructs and introduces a new military audio dataset, named MAD, which is suitable for training and evaluating audio classification systems.
- The proposed MAD dataset is extracted from various military videos and contains 8,075 sound samples from 7 classes corresponding to approximately 12 hours, exhibiting distinctive characteristics not presented in academic datasets typically used for machine learning research.
- We further conduct a comprehensive sound classification study of various deep learning algorithms on the MAD dataset.

Method

- Methodology used to collect the MAD dataset

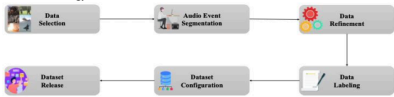


Fig. 1 Illustration of MAD dataset collection procedure.

- Data selection
 - Identify the audio content types for the seven military-related categories.
- Audio event segmentation
 - Identify the start and end times of each audio event.
- Data refinement
 - Refine the data using a variety of techniques, such as formatting, resampling, and extracting.
- Data labeling
 - Label all audio events into one of the seven predefined classes.
- Data configuration
 - Randomly split all the data into train/test sets with an approximately 9:1 ratio.

Dataset Overview

- Total Data: 8,075 audio events (12.05 hours in total)
- Dataset Split
 - Training Set: 7,393 samples (10.98 hours)
 - Test Set: 682 samples (1.07 hours)

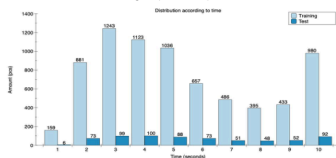


Fig. 2 Bar chart illustrating the distribution of training and test sets according to various times.

Key Features

- All audio events range from 1 to 10 seconds in duration. Meticulously labeled and categorized for high-quality deep learning-based audio classification tasks.
- Total Classes: 7 audio classes, including Communication, Gunshot, Footsteps, Shelling, Vehicle, Helicopter, and Fighter.

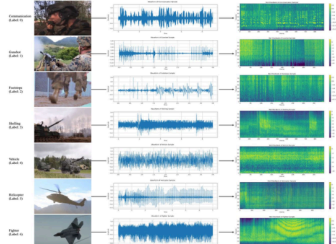


Fig. 4 The overall pre-processing task for converting waveforms to Mel filterbanks.

Technical Validation

- Model Evaluation: We conducted a comprehensive analysis of various deep learning models to evaluate the performance of the MAD dataset.
 - Models: ResNet, EfficientNet, CNN-N (PANNs), and AST (Audio Spectrogram Transformer).

- Pre-processing
 - Audio waveforms were converted into Log Mel filterbank features (80 dimensions, 25ms window, 10ms overlap).
 - All audio samples were standardized to a 10-second length using zero padding.
 - Standard normalization was applied to the spectrograms.
 - Augmentation: SpecAugment was used to mitigate overfitting.

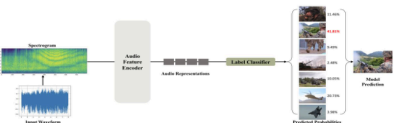
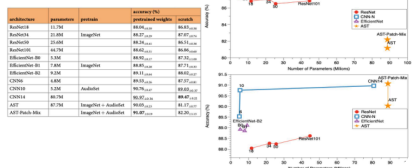


Fig. 5 Illustration of overall deep learning-based audio classification process.

Experimental Result Summary

- Pretrained models consistently outperformed those trained from scratch. With pretrained weights, the AST-Patch-Mix model achieved the highest accuracy (91.07%), aligning with the trend that models with more parameters perform better. A notable exception was the CNN10 model, which achieved a strong 89.53% accuracy with only 6% of the parameters of the AST model.
- In contrast, when trained from scratch, the CNN14 model performed best (89.47%), while the high-parameter AST models surprisingly showed the lowest performance.



03

NeurIPS 2023 DGM4H Workshop (International Conference / Co-Author)

Adversarial Fine-tuning using Generated Respiratory Sound

Tackled medical data scarcity via DiffWave diffusion synthesis, training deep classification structures with domain-adversarial fine-tuning (AFT) loops to bridge synthetic and real sample sets, increasing minority class recall by 26.58%.

- Designed medical sound synthesis pipelines using conditional audio diffusion neural vocoders, and proposed domain-adversarial fine-tuning for diagnostic label alignment.

Tech Stack: DiffWave, Domain Adversarial Training (DAT), Audio Synthesis, PyTorch



Adversarial Fine-tuning using Generated Respiratory Sound to Address Class Imbalance

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¹Kyungpook National University, ²AUI Co., Ltd, ³KAIST AI²RSC LAB, MODULABS

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Introduction

- While *deep generative models* have emerged as a promising approach in the medical image domains to address data scarcity, their use for sequential data like **respiratory sounds** is less explored.
- In this work, we propose a straightforward approach to augment imbalanced respiratory sound data using an **audio diffusion model** as a **conditional neural vocoder**.
- Nevertheless, the **discrepancy between synthetic and real samples** can introduce problems related to **distribution inconsistency**, which can degrade the performance of respiratory sound classification models as the proportion of synthetic data in the training set increases.
- We further introduce a simple yet effective **Adversarial Fine-tuning** method to **align features between the synthetic and real respiratory sound samples** to improve respiratory sound classification performance.

Method 1

- We aim to generate **high-fidelity respiratory sound samples** using audio diffusion model [1] and then **combine these synthetic samples with real data to improve the respiratory sound classification task**, especially for imbalanced lung sound disease classes.
- In phase 1, we introduce a simple method for **generating respiratory sound samples** using a **conditional neural vocoder** inspired by the recent success of *DiffWave* [1] in obtaining realistic audios.

Phase 1: Respiratory Sound Generation

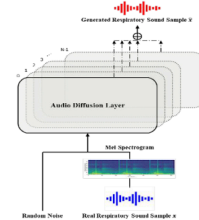


Figure 1: Overview of respiratory sound generation pipeline using audio diffusion model.

Method 2

- While augmenting real data with synthetic samples can be beneficial, we found in early experiments that in our case the **distribution mismatch between the two types of data degraded the performance of the classification model**.
- To overcome this issue, we propose a simple yet effective **Adversarial Fine-Tuning (AFT)** method inspired by [2].

Phase 2: Adversarial Fine-tuning with Mixed Samples for Respiratory Sound Classification

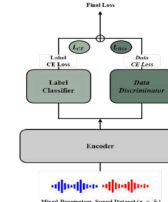


Figure 2: Overview of Adversarial fine-tuning with synthetic and real samples.

- The proposed method consists of two losses with label classifier L_C and data discriminator L_D :

$$L_C = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i), \quad L_D = -\frac{1}{N} \sum_{i=1}^N d_i \log(\hat{d}_i),$$

- where L_C and L_D are CE loss with label y and label d , and the predicted probabilities $\hat{y}$ and $\hat{d}$ are obtained by passing through the classifier and discriminator, respectively.
- To ensure that the **learned features cannot distinguish between the synthetic and real samples**, **gradients from L_D are multiplied by a negative constant during the backpropagation**.
- The final training objective is $L_{total} = L_C + \lambda L_D$, where λ is a regularization parameter drawn from [2].
- We used Audio Spectrogram Transformer (AST) [3] model as our baseline.

Results

Table 1: Performance according to various mixed sample amounts.

method	Real	Mixed 50%	Mixed 60%	Mixed 70%	Mixed 80%	Mixed 90%	Mixed 95%
AST [3]	59.55	59.92	59.92	59.92	59.92	59.92	59.92
AFT	61.79	61.79	61.79	61.79	61.79	61.79	61.79

Table 2: Accuracy (%) of the abnormal classes on the CUB dataset for AST [3] and AFT.

class	ratio	AST [3]	AFT	AST [3]	AFT
crackle (C ₁)	23.55%	45.45	42.86	41.07	46.09
wheezes (C ₂)	1.97%	36.62	36.1	22.08	27.92
both (C ₃)	5.19%	13.38	9.09	7.69	41.98

Table 3: Overall comparison of the IC-BH dataset for the respiratory sound classification task.

method	architecture	pretrain	S_C (%)	S_D (%)	Score (%)
ResNet101 [4] (CRA+BB+C+FT)	ResNet101	IN	72.30	40.10	56.30
Wang et al. [12] (CRA+BB+C+FT)	ResNet101	IN	70.40	40.20	55.30
Nguyen et al. [23] (CRA+BB+C+FT)	ResNet101	IN	70.34	37.24	58.29
Mohamed et al. [21] (CRA+BB+C+FT)	CNN	AS	75.91	39.15	67.55
Bae et al. [1] (DiffWave)	AST	IN + AS	77.14	41.97	59.55
Bae et al. [1] (DiffWave)	AST	IN + AS	81.66	43.97	62.97
AFT on Mixed-50% [ours]	AST	IN + AS	80.22	43.99	61.79

- While directly fine-tuning on mixed data did not improve the performance of the minority classes overall, **our proposed method improved their accuracies by up to 26.58%, especially in C₂**.
- Furthermore, our proposed method trained with **Mixed-50%** achieved a score of **61.79%**, **outperforming the AST [3] model by 2.24%.**

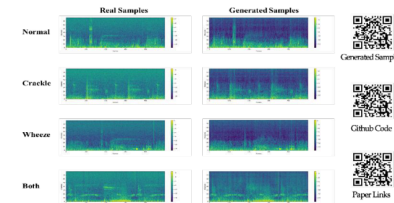


Figure 3: Comparison of spectrograms per class randomly chosen from the test set and the generated results.

References

- [1] King Derry, et al (2023), "DiffWave: A versatile Diffusion Model for Audio Synthesis", 2023.
- [2] Gao Y, et al (2019), "Domain-Adversarial Training of Neural Networks", The Journal of Machine Learning Research 2019.
- [3] Gong Y, et al (2021), "AST: Audio Spectrogram Transformer", INTERSPEECH 2021.

04

KCC Proceedings (Domestic Conference / Bronze Prize)

CutMix-SupCon: Contrastive Learning for Multimodal Emotion Recognition

Designed latent-level CutMix feature interpolation to merge KLUE-RoBERTa (text) and wav2vec 2.0 (audio) embeddings, optimizing Supervised Contrastive Learning (SupCon) for cosine similarity alignment on minor classes.

- Implemented multimodal late-fusion and contrastive learning alignments, improving speaker emotion recognition rates on class-imbalanced benchmark sets.

Tech Stack: wav2vec 2.0, KLUE-RoBERTa, Supervised Contrastive (SupCon), CutMix

CutMix-SupCon: Supervised Contrastive Learning with CutMix Augmentation for Multimodal-based Emotion Recognition

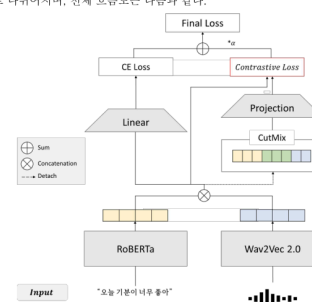
서론

- 멀티모달 데이터는 구조적으로 획득하기 어려운 문제가 있으며, 부족한 데이터 샘플의 문제로 인해 학습된 딥러닝 기반 네트워크의 실용화가 진행되기에는 여전히 한계점이 존재한다.
- 선행 연구들은 실제 환경의 부족한 데이터 문제는 고려하지 않고, 멀티모달 결합 관점에 대해서만 집중되는 경향이 존재한다.
- 본 연구에서는 멀티모달 데이터셋의 불균형한 문제를 해결하기 위해 CutMix-SupCon 방법을 제안한다. 제안된 CutMix-SupCon 방법은 잠재 공간에서 추출된 음성-텍스트 모달 간의 결합된 특징표현으로부터 혼합된 특징표현을 만들고, 혼합 대표 손실을 통해 음성-텍스트의 결합된 멀티모달 특징 표현이 원본과 얼마나 유사한지에 대한 목적 함수를 통해 최적화하였다.

방법

제안하는 멀티모달 CutMix-SupCon 방법은

- 멀티모달 특징표현 결합
 - CutMix 기반 멀티모달 혼합 증강
 - SupCon
- 으로 나뉘어지며, 전체 흐름도는 다음과 같다.



- CutMix 기반 멀티모달 혼합 증강
멀티모달 특징표현 $\mathbf{z}$ 와 레이블 y 에 대하여 샘플 $(\mathbf{x}_A, y_A)$, $(\mathbf{x}_B, y_B)$ 를 구성할 수 있고, 이로부터 새로운 샘플 $(\mathbf{z}, y)$ 를 아래의 식과 같이 만들 수 있다.

$$\mathbf{z} = M \odot \mathbf{x}_A + (1 - M) \odot \mathbf{x}_B$$

$$y = \lambda y_A + (1 - \lambda) y_B$$

- CutMix-SupCon
CutMix 기법으로부터 증강된 멀티모달 특징표현을 위한 적절한 손실 함수를 적용하기 위해, 잠재 공간내에서 혼합된 정보를 구분하여 효과적인 학습을 가능하게 하는 CutMix-SupCon 방법을 제안한다.

$$L_{CL} = \sum_{i \in I} \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\mathbf{z}_i \cdot \frac{\mathbf{z}_p}{\|\mathbf{z}_p\|})}{\sum_{q \in A(i)} \exp(\mathbf{z}_i \cdot \frac{\mathbf{z}_q}{\|\mathbf{z}_q\|})}$$

$$L = L_{CE} + \alpha L_{CL}$$

위와 같은 최종 손실 함수를 통해 모델은 혼합된 멀티모달 특징표현이 원본의 샘플의 고유 정보를 얼마나 포함하는지를 학습한다.

실험 및 결과

- 데이터셋: KEMDy19 Text+Audio 멀티모달 감정인식 데이터셋
중복된 파일, 음성의 발화 세그먼트, 20초 이상의 음성 파일 제거 등의 전처리를 통해 10,214개의 데이터 샘플을 확보
학습 및 평가셋 8:2 무작위 분할
- 방법: 각 모달에 대한 특징표현 기반의 CE, Mixup, CutMix, 제안하는 CutMix-SupCon 방법에 대한 결과 보고

Input	Method	Ace (%)	F1-score (%)
Text	Cross-Entropy	73.42	69.32
Audio		72.59	67.13
Text+Audio	Mixup	73.57	69.08
Text		71.76	67.37
Audio	CutMix	71.46	65.89
Text+Audio		72.39	68.94
Text	CutMix-SupCon (ours)	72.93	69.37
Audio		71.76	65.94
Text+Audio	CutMix-SupCon (ours)	74.74	70.77
Text		72.49	69.72
Audio	CutMix-SupCon (ours)	72.54	68.27
Text+Audio		75.28	71.17

표 1 KEMDy19 데이터셋을 사용한 감정인식 평가 결과

Input	Method	Weighted F1-score(%)
Text+Audio	Cross-Entropy	73.55
	Mixup	72.39
	CutMix	74.77
	CutMix-SupCon (ours)	74.96

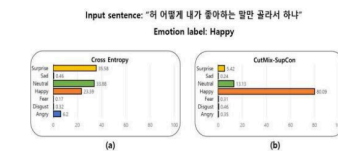


그림 2 KEMDy19 테스트 데이터 샘플에 대한 CE와 CutMix-SupCon의 예측 확률 분포 비교